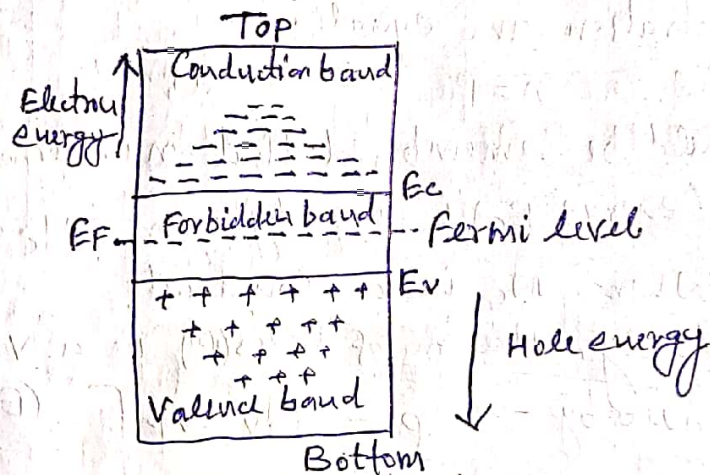


Fermi Level :- Fermi level is a characteristic energy of the material.

In intrinsic semiconductors, there is an equal no of electrons and holes. The concentration of electrons decreases above the bottom of conduction band and similarly the concentration of holes decreases below the top of the valence band as shown in fig below. This fig shows that the centre of gravity of electrons and holes lies exactly at the middle of forbidden gap. The central level is known as Fermi level.



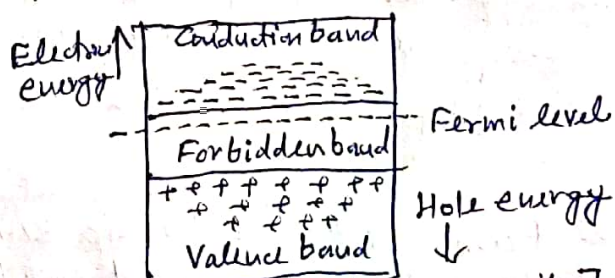
(Fig: Showing Fermi level)

→ Thus, Fermi level is the energy that corresponds to the centre of gravity of the conduction electrons and holes weighed according to their energies.

Fermi level is a concept like as a hollow body can have a centre of gravity where there is no matter.

→ When a donor impurity is added to an intrinsic semiconductor, it becomes N-type. Now it has more conduction electrons than holes as shown in fig below.

i.e. Fermi level shifts towards the conduction band.

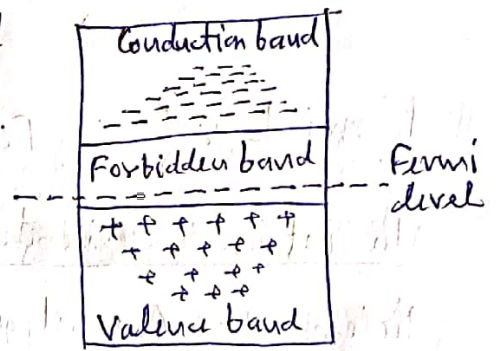


[Fig: Fermi level with donor impurity]

Similarly, when an acceptor impurity is added to an intrinsic semiconductor it becomes P-type.

Now it has more holes than electrons.

This shifts the Fermi level towards the Valence band as shown in fig.



[Fig: Fermi level with acceptor impurity]

Position of Fermi level in Intrinsic Semiconductors :-

In an intrinsic semiconductor, the free electron and hole concentration are equal.

$$\text{i.e. } n = p \quad (1)$$

it should be remembered that number of electrons (n) in conduction band (energy E_c) is given by - $n = N_c \exp \left[-\frac{(E_c - E_F)}{KT} \right]$ — (2)

Where N_c is a constant.

Similarly, the no of holes (p) in a valence band (with energy E_v) is given by - $p = N_v \exp \left[-\frac{(E_F - E_v)}{KT} \right]$ — (3)

Substituting the values of n and p from eqs (2) & (3) in eq (1), we get - $N_c \exp \left[-\frac{(E_c - E_F)}{KT} \right] = N_v \exp \left[-\frac{(E_F - E_v)}{KT} \right]$

$$\therefore \frac{N_c}{N_v} = \frac{e^{-\frac{(E_F - E_v)}{KT}}}{e^{-\frac{(E_c - E_F)}{KT}}} \quad (4)$$

Taking logarithm of both sides, we have -

$$\log \left(\frac{N_c}{N_v} \right) = -\frac{(E_F - E_v)}{KT} + \frac{(E_c - E_F)}{KT}$$

$$\text{or } \log \left(\frac{N_c}{N_v} \right) = \frac{E_c + E_v - 2E_F}{KT} \Rightarrow E_F = \frac{E_c + E_v}{2} - \frac{KT}{2} \log \frac{N_c}{N_v} \quad (5)$$

if the effective mass of electron is equal to the effective mass of hole

i.e. $m_n = m_p$, then $N_c = N_v$ and eq (5) becomes -

$$E_F = \frac{E_c + E_v}{2} \quad (6)$$

Thus, Fermi level in intrinsic semiconductor lies exactly halfway between valence band and conduction band.

Fermi level in Extrinsic Semiconductor :-

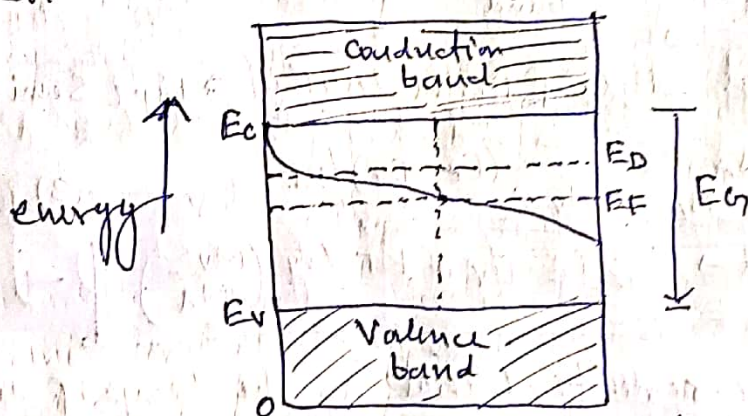
In Case of an intrinsic Semiconductor, the Fermi level E_F lies in the middle of energy gap. This shows that free electrons and holes have equal concentrations.

Now Let us see how the Fermi level changes in an extrinsic s/c.

(1) Fermi level in N-type extrinsic Semiconductor :-

Fermi level is a measure of the probability of occupancy of the allowed energy states.

Hence the Fermi level must move closer to the conduction band to indicate that many energy states in that band are filled by the donor electrons and fewer holes exist in valence band. This situation is shown in fig below:



[Fig: Position of Fermi level in N-Type Semiconductor]

As N-type Semiconductor is almost entirely due to extrinsically supplied electrons from the donors and hence $n = N_D$ where N_D is the concentration of donor atoms.

Substituting $n = N_D$ we have - $n \approx N_D = N_C e^{-(E_C - E_F)/KT}$ (1)

Taking logarithm, we have - $\log N_D = \log N_C - (E_C - E_F)/KT$

Solving we get - $E_F = E_C - KT \log (N_C/N_D)$ (2)

(2) Fermi level in P-type Extrinsic Semiconductor :-

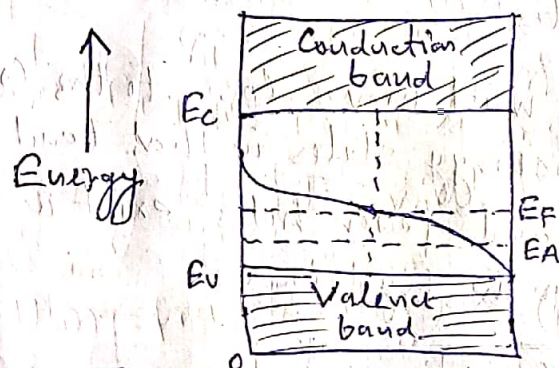
Here it is assumed that :-

(i) The density of acceptor atoms far exceeds the density of donor atoms.

(ii) Every acceptor atom has accepted one electron from the valence band.

(iii) The density of electrons in the conduction band is much smaller than the density of holes in the valence band.

The same kind of argument (as in N-type extrinsic s/c) leads to the conclusion that the Fermi level must move from the centre of forbidden gap closer to the valence band for a P-type material. The situation is shown in fig below -



[Fig: Position of Fermi level in P type Semiconductor]

In this case $n = N_A$
 $\therefore n \cong N_A = N_V e^{-(E_F - E_V)/KT}$ — (3)

Solving we get $\rightarrow E_F = E_V + KT \log \left(\frac{N_V}{N_A} \right)$ — (4)

Here it should be remembered that as the temp^r of either N-type or P-type material increases, the Fermi level moves towards the centre of the energy gap.

Ques-1:- In an N-type semiconductor, the Fermi level is 0.3 eV below the conduction band at 300K. if the temp^r is increased at 330K find the new position of Fermi level.

Solⁿ:- We know that $E_F = E_C - KT \log \left(\frac{N_C}{N_D} \right) \Rightarrow E_C - E_F = KT \log \left(\frac{N_C}{N_D} \right)$

Given $E_C - E_F = 0.3 \text{ eV}$ when $T = 300 \text{ K}$

Therefore, $0.3 = 300 \text{ K} \log \left(\frac{N_C}{N_D} \right)$ — (1)

Let when the temp^r is 330 K, E_{F1} be the new position of Fermi level.

In this case $E_C - E_{F1} = 330 \text{ K} \log \left(\frac{N_C}{N_D} \right)$ — (2)

Dividing eqⁿ (2) by eqⁿ (1) we get $\frac{E_C - E_{F1}}{0.3} = \frac{330}{300} = 1.1 \Rightarrow E_C - E_{F1} = 0.3 \times 1.1 = 0.33 \text{ eV}$

Ques-2:- At what temp^r can we expect a 10% probability that ⁽⁵⁾ electrons in a metal will have an energy which is 1% above E_F . The Fermi energy of the metal is 5.5 eV.

Solⁿ:- Given $f(E) = 10\% = 0.1$ and $E_F = 5.5 \text{ eV}$

$$\text{Now } E = E_F + 1\% \text{ of } E_F = 5.5 + \frac{5.5}{100} = 5.5 + 0.555$$

$$\therefore E - E_F = (5.5 + 0.555) - 5.5 = 0.555$$

$$\text{We know that- } f(E) = \frac{1}{\exp\left[\frac{E - E_F}{KT}\right] + 1}$$

$$\therefore 0.1 = \frac{1}{\exp\left[\frac{0.555 \times (1.6 \times 10^{19})}{(1.38 \times 10^{23}) T}\right] + 1} = \frac{1}{\exp\left(\frac{637.7}{T}\right) + 1}$$

$$\text{Solving we get } \boxed{T = 290 \text{ K.}}$$

Ques-3:- Using free electron model, find the Hall Coefficient of Sodium assuming BCC structure for Na of Cell Side $4.28 \text{ \AA}$.

Solⁿ:- Unit Cell of Sodium atom (Na) of volume a^3 has 2 atoms.

$$\text{Therefore- } n = 2 \times \left(\frac{1}{a^3}\right) = 2 \times \left[\frac{1}{(4.28 \times 10^{-10})^3}\right] = 2.55 \times 10^{28} / \text{m}^3$$

$$\text{Now, Hall Coefficient } R_H = -\frac{1}{ne}$$

$$\therefore R_H = \frac{1}{(2.55 \times 10^{28} \times 1.6 \times 10^{19})} = -0.245 \times 10^{-9} \text{ m}^3/\text{C}$$

Ques-4:- The resistivity of a sample semiconductor is 9 milli-ohm-metre its holes have mobility of $0.03 \text{ m}^2/\text{Vs}$, Calculate Hall Coefficient

Solⁿ:- We know that-

$$\text{Conductivity, } \sigma = \frac{1}{\text{resistivity}} = \frac{1}{9 \times 10^{-3}} = 111.11 \text{ } \Omega^{-1}\text{m}$$

$$\text{Now, Hall Coefficient, } R_H = \frac{\mu}{\sigma} = \frac{0.03}{111.11} = 2.7 \times 10^{-4} \text{ m}^3/\text{C}$$

$$\boxed{R_H = 2.7 \times 10^{-4} \text{ m}^3/\text{C}}$$

Ques-5 :- Hall voltage of 1.0 mV is found to be developed when a sample carrying a current of 10.0 mA is placed in a transverse magnetic field of 0.3 T . Calculate the Charge carrier concentration of the sample, given the thickness of the sample along the direction of magnetic field is 0.3 mm .

Soln :- We know, that -

$$R_H = \frac{V_H Z}{I_x B_z} \quad \text{and} \quad R_H = \frac{1}{ne}$$

$$\therefore \frac{1}{ne} = \frac{V_H Z}{I_x B_z} \quad \text{or} \quad n = \frac{I_x B_z}{V_H Z e}$$

Substituting the given values, we get -

$$n = \frac{(10 \times 10^{-3}) \times (0.3)}{(1 \times 10^{-3}) \times (0.3 \times 10^{-3}) \times (1.6 \times 10^{-19})}$$

$$= \frac{3 \times 10^{-3}}{0.3 \times 10^{-6} \times 1.6 \times 10^{-19}}$$

$$= 6.25 \times 10^{22} \text{ m}^{-3}$$

$$\boxed{n = 6.25 \times 10^{22} / \text{m}^3}$$